



Team Kappa

Intermediate Maze—Brisbane Boys' College

The Team

Aryan - Programmer

Chris - Electrical

Rory - Design

Elliott - Generalist

Fun Fact:

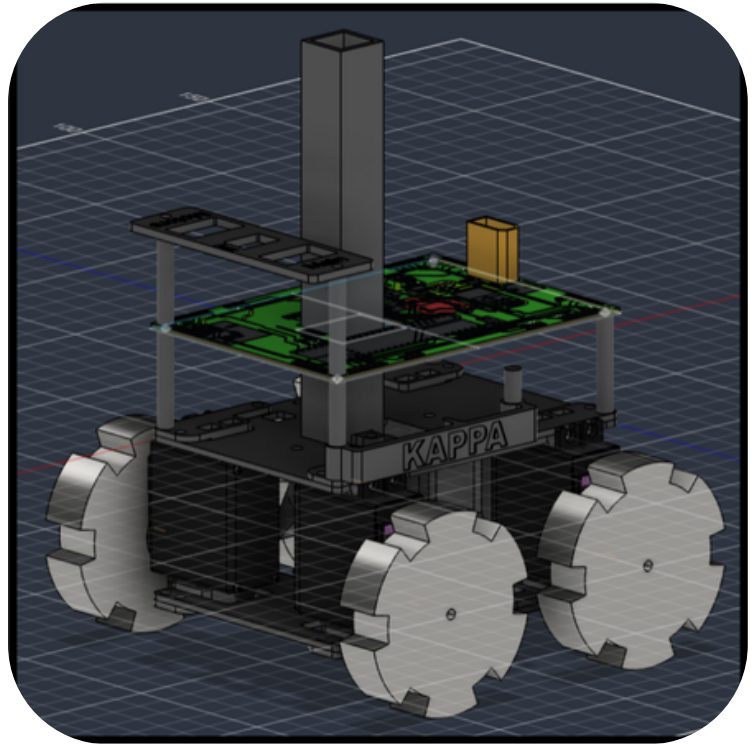
It no longer takes hours to disassemble our robot!

Challenges Endured:

We underestimated the time needed to master electronics, C++ coding, and design for manufacturing. Through determination and focus, we pulled everything together in time for the competition. Early on, we also created overly complex designs; we resolved this by bringing in teammates from other areas to offer a fresh perspective and streamline our approach.

Learning curves of the project:

- . Trouble shooting centering
- . 3D modeling the dropper
- . Making all parts fit
- . Bringing together the cohesion of the project



Finite State Machine code example:

```
void loop() {  
    // FSMs  
    switch (state) {  
        case FORWARD:  
            forward();  
            break;  
        case ROTATE_L:  
            rotate_left();  
            break;  
        case ROTATE_R:  
            rotate_right();  
            break;  
        case ROTATE_180:  
            rotate_180();  
            break;  
        case NAV:  
            navigation();  
            break;  
        case BT_BACK:  
            black_tile_backwards();  
            break;  
        case BT_ROTATE:  
            black_tile_rotate();  
            break;  
        case VICTIMS:  
            victims();  
            break;  
        case SILVER:  
            silver_tile();  
            break;  
        case PAUSE:  
            pause();  
            break;  
        default:  
            state = NAV;  
            break;  
    }  
}
```

Lessons Learnt in 2026:

Through the development process we have learnt key take aways. We severely underestimated the amount of time it would take to code, meaning we couldn't refine it as much as we wanted. To spread this load, specialists in our group often helped out in other areas of development to get a second set of eyes to check when needed. Furthermore we created a plan so the group would not get fixated on one individual thing or become off track.